

1 Data Preprocessing in Retail Analytics

A. Handling Missing Values

1. Missing values:

- * Age $\rightarrow$ C2

- * Monthly-Spend $\rightarrow$ C3

- * Purchase-Frequency $\rightarrow$ C5

2. Imputation methods:

- * Median Imputation

- * K-NN Imputation

3. Why?

- * Median is good because it is not affected by outliers.

- * k-NN uses similar records to fill missing values.

B. Outlier Detection and Treatment

1. Outliers:

- * Age = 105

- * Monthly-Speed = 500

2. Records:

- * C4

3. Treatment:

- * Cap the values or remove the record if necessary.

C. Categorical Data Standardization

- * M, Male, male $\rightarrow$ Male

- * F, female, FEMALE $\rightarrow$ Female

2 Covariance & Data Reduction

A. Covariance Analysis

1. Income and Purchase-Value have positive covariance.
2. As income increases, purchase value also increases.
3. Covariance shows the relationship between two features.

B. Feature Selection

1. Remove redundant features like Income and Avg-Balance if highly correlated.
2. Keep useful features: Age, Income, Credit-Score, Orders.

C. Data Transformation

1. Standardize all numeric attributes.
2. It makes all features have the same scale and improves model performance.

3 Data Transformation & Discretization

A. Equal-Width Binning

- * Low
- * Medium
- * High

B. Equal-Frequency Binning

Divide data into 3 groups with nearly equal records.

C. Concept Hierarchy

- * Study Hours $\rightarrow$ Low $\rightarrow$ Medium $\rightarrow$ High
- * Test Score $\rightarrow$ Poor $\rightarrow$ Average $\rightarrow$ Excellent

4 Normalization

a) Min-Max Normalization Formula!

$$X' = \frac{X - X_{\min}}{X_{\max} - X_{\min}}$$

b) Z-Score Normalization Formula!

$$Z = \frac{X - \mu}{\sigma}$$

5 Online shopping Behavior Analysis

A. Equal-Frequency Binning

- * Divide visit durations into 3 equal groups.
- * Each group has nearly the same number of records.

B. Z-score Normalization

- * Find mean and standard deviation.
- * Convert values using the z-score formula.
- * Extreme values become easier to identify.

C. Min-Max Normalization

- * Convert all values between 0 and 1.
- * Easy to compare values but sensitive to outliers.